FLOCK ENERGY | DATA SCIENCE TAKE-HOME

Washing-Machine Load Disaggregation from REFIT Smart-Meter Data

Data quality, exploratory analysis, household-separated evaluation and practical implications

Technical report | Analysis period: 1 September - 30 November 2014 (UTC)

Data: REFIT original raw House 1 and official cleaned household readings. **Model:** Random Forest with a zero-consumption baseline and validation-selected post-processing.

Repository: github.com/sourav349/refit-energy-disaggregation-ml

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Executive finding

A reproducible pipeline was built to clean one-minute electricity readings, examine washing-machine usage across four UK homes, and estimate washer power from whole-house measurements. The model does recognise some operating periods, but it cannot yet distinguish the washing machine reliably from other household electrical loads. On the final House 6 holdout, the validation-selected Random Forest achieved an active-minute F1-score of **0.303**, but its overall mean absolute error was **5.39 W**, compared with **3.12 W** for the always-off baseline. It should therefore be regarded as a research proof of concept, not a deployable appliance-monitoring product.

Evidence note: All numerical findings and figures in this document are drawn from the previously executed project outputs supplied for this report. The initial House 5 evaluation and subsequent House 6 holdout are distinguished throughout.

1. Problem, scope and contribution

A smart meter measures the total electricity consumed by a home, not the individual appliances responsible. When a washing machine turns on, its electrical demand is mixed with that of refrigeration, lighting, cooking equipment and other devices. The task is to estimate the washing-machine component from the aggregate reading alone. This is a form of non-intrusive load monitoring (NILM).

The assignment has four linked objectives: inspect and clean raw House 1 data; describe washing-machine behaviour across at least four homes; develop and evaluate a simple appliance-disaggregation model; and translate the findings into a realistic utility recommendation. The emphasis is on transparent decisions and reproducibility rather than architectural complexity.

Why a zero baseline is necessary

Most one-minute observations occur while a washing machine is off. A model that always returns zero can therefore obtain a low overall error despite detecting no active washing-machine periods. It is essential to compare any learned model against this baseline and to report detection metrics and errors restricted to genuinely active minutes.

Scope and reproducibility

The reported analysis uses minute-averaged power readings for a requested window of 1 September to 1 December 2014 (end exclusive, UTC). Raw data are used for the cleaning audit, and official cleaned data are used for cross-household analysis and model training. Data archives are downloaded separately and are not committed to the public repository.

2. Dataset and study design

REFIT contains electricity measurements from 20 UK households, with source readings sampled at approximately eight-second intervals. Each home includes whole-house aggregate power and individual appliance channels, measured in watts (W). Appliance-column assignments differ between homes, and their coverage and completeness are not identical.

House	Washing-machine channel	Role
1	Appliance5	Raw cleaning; model training
2	Appliance2	Model training
3	Appliance6	Validation and model selection
5	Appliance3	Initial test and error analysis
6	Appliance2	Later holdout after model selection

It is important not to assume that the same appliance column represents the washer in every home. Model inputs were derived from aggregate readings and their histories; the appliance channel was used as a supervised-learning target and for evaluation only.

3. Raw data audit and cleaning

The raw House 1 series was inspected for timestamp parsing, ordering, duplicate records, irregular sampling, missing measurements and nonphysical power values. Duplicate UTC timestamps were resolved using a documented last-observation rule, followed by chronological sorting. Negative values were invalidated. Implausibly high appliance values above 4,000 W were treated as invalid, while aggregate readings above 25,000 W were flagged for review rather than removed automatically.

Measured raw-data quality

Audit indicator	Recorded finding
Raw House 1 input records	1,400,757
Duplicate timestamps	170,622
Unparseable timestamps	0
Missing one-minute bins	3,334

Raw readings were aggregated to one-minute means, because the arithmetic mean estimates power during each minute. It would be incorrect to sum power readings and treat the result as energy. For an observed one-minute interval, estimated energy in kWh is mean power in W divided by 60,000.

Handling outages

Short missing runs of one or two whole minutes were filled only when bounded by valid observations. Longer outages remained missing rather than being converted to zero. In the official cleaned dataset, some missing sections have already been imputed, so apparent zero usage should not always be interpreted as confirmed appliance inactivity. Where present, the REFIT issue flag was used to exclude inconsistent readings.

The first figure in the appendix shows the same House 1 signals before and after this processing. The most important consequence is a consistent one-minute time axis, not the elimination of every high-power spike.

4. Washing-machine usage exploration

To identify candidate washing-machine operating events, a simple heuristic marked the appliance active when observed power exceeded 20 W. Pauses of up to 10 minutes were joined only when intervening observations existed. An event was retained if its total span was at least 15 minutes and it included at least three active one-minute samples. This is an operational rule, not manufacturer-verified cycle labelling.

House	Observed minutes	Missing minutes	Candidate events	Median kWh/event
1	124,390	5,635	17	0.0399
2	120,576	10,464	99	0.0704
3	121,682	8,343	77	0.0653
5	117,080	13,960	131	0.0801

House 5 generated the greatest number of candidate events, while House 1 generated the fewest. This is not sufficient to conclude that one household uses more energy overall: sensor coverage, missingness, differing appliance behaviour and the heuristic event definition all affect the counts. The plots of cycle duration, usage timing and energy should be interpreted as exploratory rather than verified laundry diaries.

5. Model development and leakage control

The baseline predicted zero washing-machine watts at every observed minute. The learned model was a RandomForestRegressor trained on whole-house aggregate power and causal engineered features: lagged observations at one, five and fifteen minutes; trailing means and standard deviations; short-term aggregate changes; and encoded hour and weekday. Predictions were constrained to physically plausible nonnegative values not exceeding the observed aggregate.

The core model was fitted on Houses 1 and 2. House 3 was used for validation and model selection. House 5 served as the initial unseen-house evaluation, but its errors were subsequently inspected; it must therefore not be represented as untouched for later model-development decisions. The final selected configuration was evaluated separately on House 6 after validation decisions were locked.

Why split by household?

Randomly dividing adjacent one-minute records into training and testing sets would mix neighbouring samples from the same appliance operation across both partitions. This can give an optimistic estimate of generalisation. Separating entire households provides a more meaningful measure of transfer to an unseen home, although four or five households cannot establish performance across a larger utility population.

Model evaluation measures

Overall MAE quantifies average absolute watt-level error. **Active-minute MAE** evaluates errors specifically where the washer was truly running. **Precision** is the fraction of predicted-active minutes that were actually active, and **recall** is the fraction of truly active minutes detected. **F1** balances precision and recall. **Energy bias** compares total estimated energy with measured energy; its sign shows over- or underestimation.

Initial validation and House 5 evaluation

Split	Method	Overall MAE (W)	Active F1	Energy bias
H3 validation	Always off	4.37	0.000	-100.0%
H3 validation	Random Forest	14.29	0.205	+180.9%
H5 initial test	Always off	10.46	0.000	-100.0%
H5 initial test	Random Forest	18.20	0.148	+7.38%

The Random Forest detected some activity but had a higher overall error than the always-off baseline in both homes. That is a meaningful negative result: the learned model had not produced a reliably useful general-purpose washer estimate.

6. Error analysis and follow-up experiments

Error inspection on House 5 revealed both false positives and missed genuine activity. At 06:03 UTC on 30 November, the aggregate reading was about 2,449 W and measured washer load was 0 W, yet the model predicted approximately 1,464 W. At 09:11 UTC on 14 November, measured washer load was about 2,092 W, while the model predicted only 3.8 W. A similar aggregate load can therefore arise from quite different appliances; power magnitude alone does not identify a washer.

Calendar features and output thresholds

The original feature-importance ranking assigned substantial weight to time-of-day and day-of-week encodings. Removing those variables reduced validation overall MAE from 14.29 W to 12.69 W but also lowered active F1 from 0.205 to 0.163. A 50 W minimum prediction cutoff applied to the original model produced the best F1 of these tested variants, approximately 0.216 on House 3.

House 3 candidate	MAE (W)	Precision	Recall	F1
Original RF	14.29	0.119	0.727	0.205
RF without calendar	12.69	0.097	0.504	0.163
Original RF, 50 W cutoff	11.77	0.144	0.427	0.216
No calendar, 50 W cutoff	10.53	0.103	0.275	0.150

Two-stage model: tested and rejected

A second experiment first classified active versus inactive minutes, then estimated wattage during predicted activity. Across the tested classifier-probability thresholds, the two-stage approach produced substantially more false positives and did not beat the original Random Forest with the 50 W cutoff on House 3 active F1. It was therefore rejected. This matters because a more elaborate model is not necessarily a better model.

House 3 was inspected repeatedly during this work, so its validation performance may be optimistic. The final House 6 evaluation below is reported separately without subsequent tuning.

7. Later independent holdout: House 6

After model selection on House 3, the original Random Forest with the 50 W output cutoff was evaluated on House 6, whose washing-machine channel is Appliance2. The supplied experiment report records 93,695 model-ready minutes within the selected observation window.

House 6 method	Overall MAE	Active MAE	Precision	Recall	F1	Energy bias
Always off	3.12 W	319.20 W	0	0	0	-100%
Selected RF + 50 W cutoff	5.39 W	237.75 W	0.217	0.500	0.303	+27.45%

The selected method detected half of the genuinely active minutes, but only 21.7% of predicted-active minutes were correct. Its active-minute MAE improved relative to always-off by about 81.45 W, yet overall MAE was worse by 2.27 W. Total estimated washing-machine energy was 27.45% higher than the measured total.

This mixed outcome is consistent with the task difficulty: a model can provide partial information about usage events while also producing enough false alarms to increase its error across the full day. The observed F1-score is not adequate evidence for reliable customer-facing claims about individual washing-machine cycles.

What the House 6 plot adds

The House 6 trace in Figure 7 shows missed high-power events and apparent false positives during one illustrative day. It is a diagnostic example, not an additional performance test. The numerical results above apply to the full set of model-ready House 6 observations.

8. Practical utility recommendation

The most credible near-term opportunity is an **opt-in usage-awareness pilot** rather than automated appliance-specific billing or definitive washing-machine notifications. If a customer has a suitable time-of-use tariff, the utility could test whether carefully qualified insights about likely laundry timing encourage voluntary load shifting into lower-priced or less congested periods.

The pilot should first validate cycle detection against appliance submeters. A controlled evaluation could then measure estimated and verified peak-period kWh shifted, changes in customer bills, false notification rates, participation, opt-outs and customer feedback. Shifting electricity consumption to another time does not necessarily reduce total kWh; any savings claim must distinguish bill savings from energy savings.

9. Limitations, next steps and responsible interpretation

Limitations of data and model

- Overlapping signatures: kettles, heaters, cookers and other appliances can create changes similar to washer heating cycles.
- Household transfer: machine types, wash programmes and household routines differ between training and evaluation homes.
- Imputation and sensor quality: the official cleaned dataset may include filled zeros and quality flags, so not every apparent off state represents verified absence of use.
- Cycle boundaries are heuristic: thresholds and pause rules alter event counts, durations and energy summaries.
- At 15- or 30-minute resolution, short heating peaks and event boundaries are smoothed or lost; one-minute metrics cannot be reused without new validation.
- The REFIT homes are in the UK. Transfer to Indian houses requires new regional meter readings, appliance patterns, tariff context and field validation.

Reasonable improvements

Further work should focus on independent household evaluation, more reliable training labels, calibration, sensitivity analyses for the event-detection thresholds, and sequential features that distinguish sustained appliance behaviour from isolated high-power bursts. A broader cross-household study would be more valuable than repeatedly tuning against the same small holdout set.

Development and AI-tool disclosure

Python, Pandas, scikit-learn and Matplotlib were used for processing, modelling and visualization. ChatGPT assisted with project scaffolding, code suggestions and documentation. The recorded REFIT runs, actual numerical outputs and visual plots must remain the basis for claims; any further experiment should be reproducible from the public repository.

10. Reproduce the analysis

The source code and required documentation are published in the project repository. The REFIT archives are not included because of their size. After extracting the two official archives into appropriate local folders, install the requirements and run the processing pipeline. For the original local folder arrangement, the command is:

python -m refit_nilm.pipeline --raw-dir REFIT_RAW_081116 --clean-dir CLEAN_REFIT_081116 --output-dir outputs --start 2014-09-01 --end 2014-12-01 --max-train-rows 120000

Reproduction code, measured data-cleaning observations, model results and the one-page utility-facing recommendation are described in **README.md**, **DATA.md**, **RESULTS.md** and **Recommendation.md**, respectively. The GitHub submission is the primary executable deliverable; this PDF is its companion report.

Open the public GitHub repository

References and data attribution

University of Strathclyde. *REFIT: Electrical Load Measurements* (raw). Dataset page.

University of Strathclyde. *REFIT: Electrical Load Measurements (Cleaned)*. Cleaned dataset page.

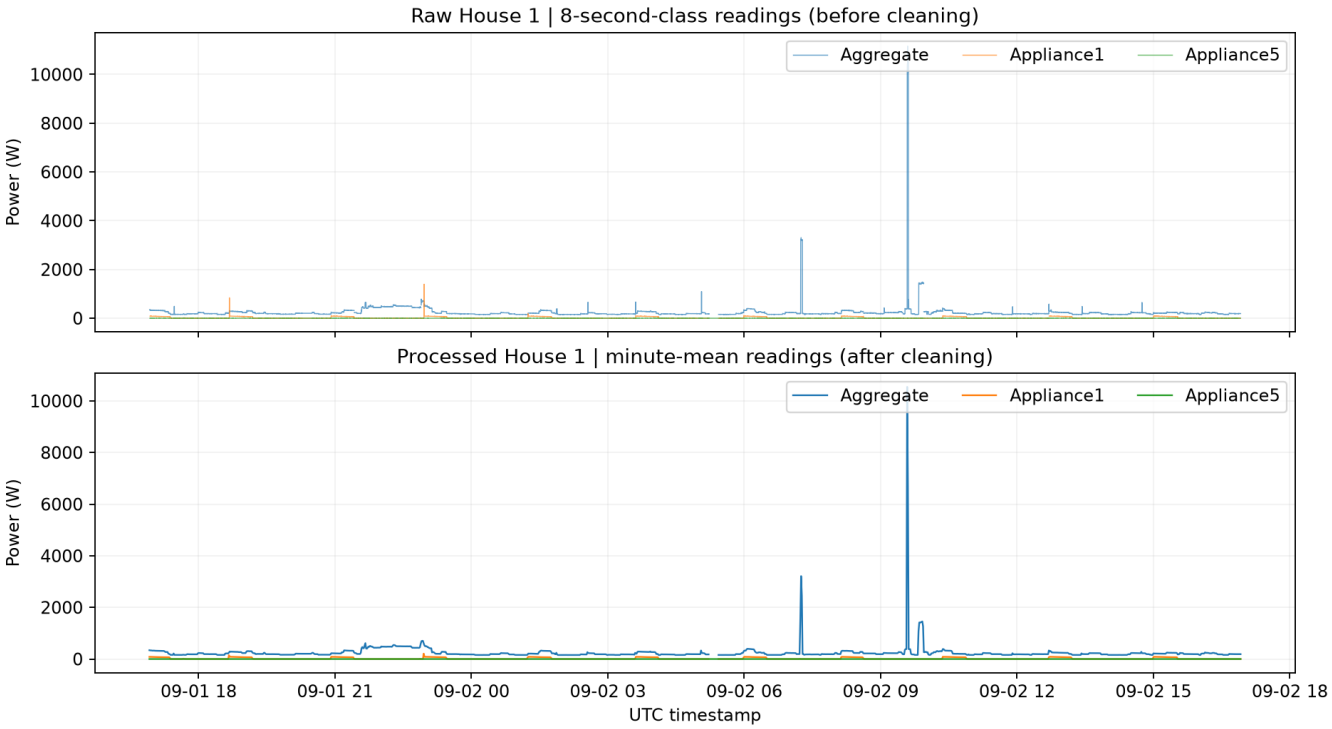
University of Strathclyde. *Cleaned REFIT README*. Sensor mapping and cleaning notes.

Murray, D., Stankovic, L. and Stankovic, V. (2017). An electrical load measurements dataset of United Kingdom households from a two-year longitudinal study. *Scientific Data*, 4, 160122. https://doi.org/10.1038/sdata.2016.122.

Dataset attribution: Creative Commons Attribution 4.0 (CC BY 4.0), subject to the terms on the official data portal.

The original graphics supplied with this request are reproduced without changing their numerical content. Their interpretations should be read together with the caveats and split definitions in Sections 4-9.

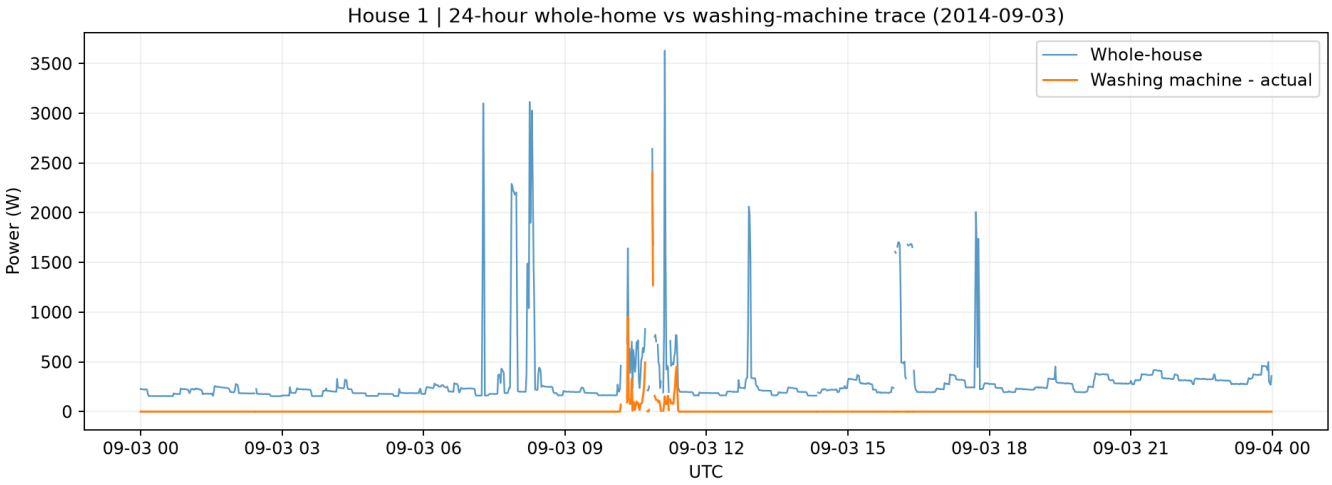
Figure 1. Raw and processed House 1 readings



Source: *Original plot supplied from the executed REFIT project. Raw and processed House 1 readings.*

Interpretation. Minute averaging produces a consistent sampling interval and reduces short fluctuations. The large aggregate spikes remain visible; cleaning should not automatically discard physically possible usage peaks.

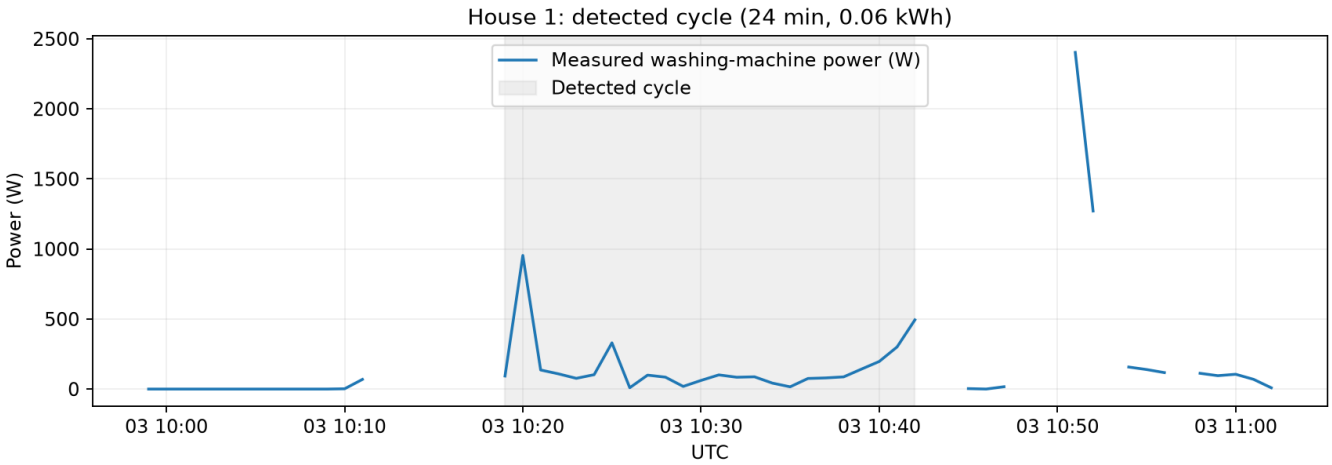
Figure 2. Whole-home and washing-machine power, House 1



Source: *Original plot supplied from the executed REFIT project. Whole-home and washing-machine power, House 1.*

Interpretation. The washer signal is present during a limited part of the day, while many aggregate spikes occur at other times. That is why aggregate peak size alone is an unreliable washer detector.

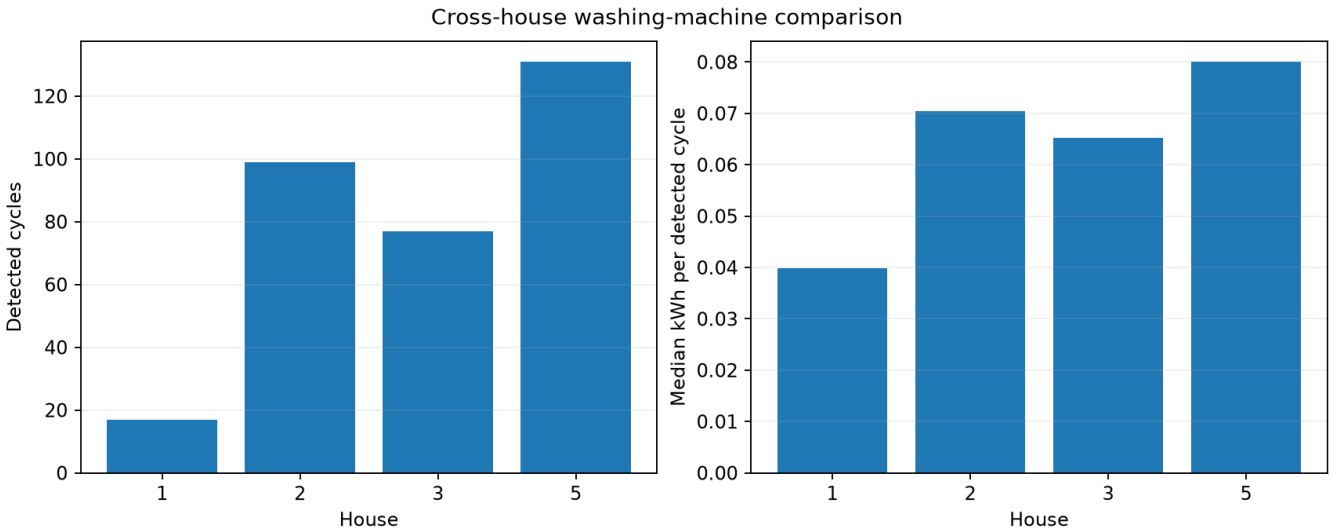
Figure 3. Example washing-machine candidate cycle



Source: *Original plot supplied from the executed REFIT project. Example washing-machine candidate cycle.*

Interpretation. The shaded 24-minute interval is identified through the documented >20 W activation and pause-merging rule. It is an inferred event, not an independently labelled complete wash programme.

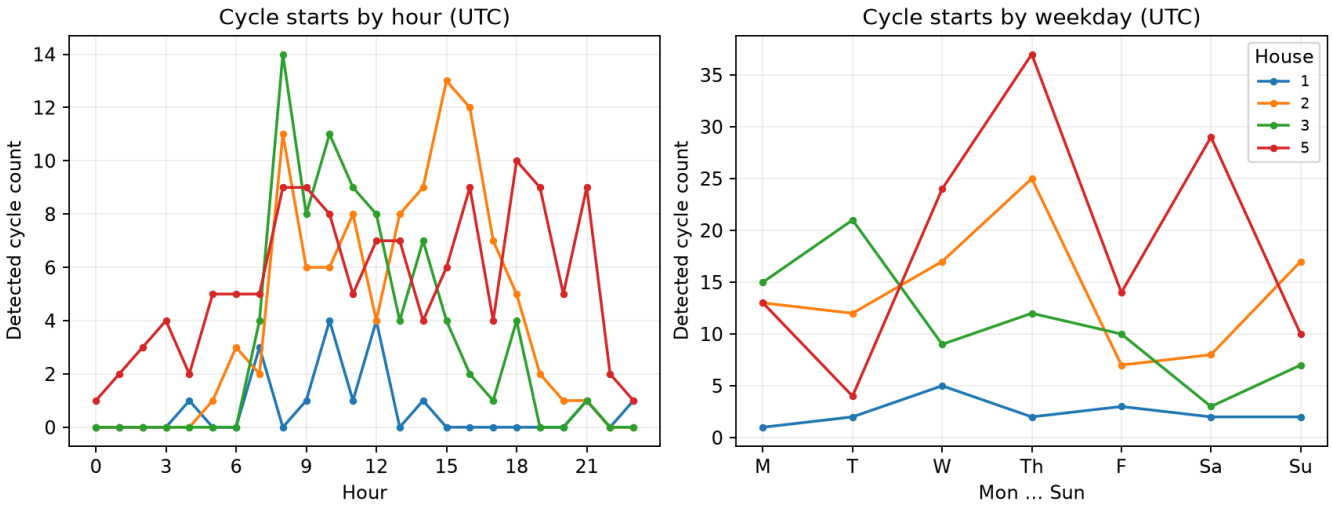
Figure 4. Cross-house candidate events and energy per event



Source: *Original plot supplied from the executed REFIT project. Cross-house candidate events and energy per event.*

Interpretation. Detected event counts vary considerably between homes. Comparison of raw counts alone ignores coverage and possible differences in appliances; median event energy supplies complementary context.

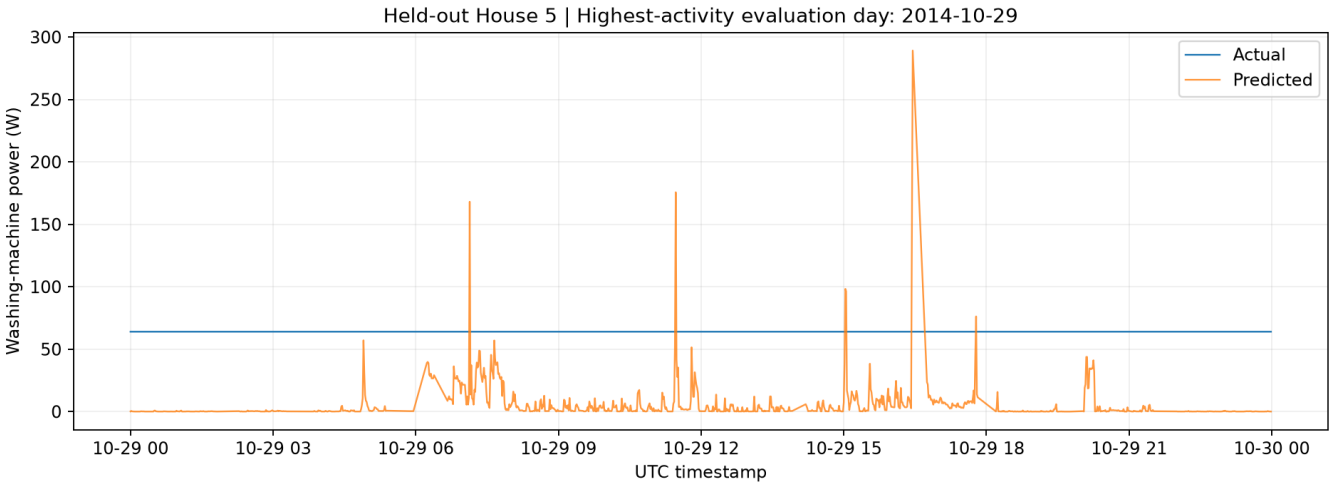
Figure 5. Candidate cycle starts by hour and weekday



Source: *Original plot supplied from the executed REFIT project. Candidate cycle starts by hour and weekday.*

Interpretation. The graphs reveal household-specific temporal patterns, but these timestamps are UTC rather than local British time. Different event counts and missingness also influence the curves.

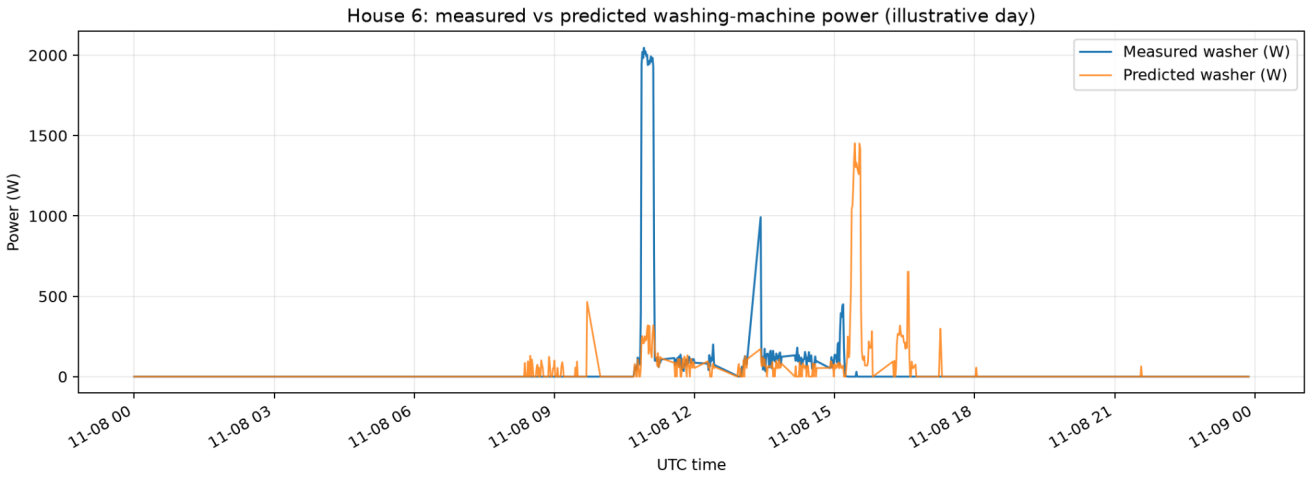
Figure 6. Initial House 5 actual versus predicted washer power



Source: *Original plot supplied from the executed REFIT project. Initial House 5 actual versus predicted washer power.*

Interpretation. Predicted spikes and the nearly flat measured trace suggest a poor visual match. The unusually flat measured line should be checked against the underlying test_predictions.csv before using this chart as strong evidence. The full-period metrics remain the primary evaluation.

Figure 7. Final House 6 holdout: measured versus predicted power



Source: *Original plot supplied from the executed REFIT project. Final House 6 holdout: measured versus predicted power.*

Interpretation. The selected model misses some high measured peaks and produces other substantial estimates without corresponding measured activity. The full holdout evaluation is summarised in Section 7.